

Cashew Disease Detection

State-of-the-Art XAI & Hybrid Modeling

Achievement: 99.02% Accuracy

Architecture: EfficientNet-B0 + CaiT Transformer

Explainable AI: Grad-CAM and SHAP Integrated

1. Abstract & Introduction

This research presents a robust deep learning framework for the automated identification of cashew leaf diseases. By leveraging a hybrid architecture that integrates convolutional neural networks (CNN) for local feature extraction and vision transformers for global context, we achieve human-expert level classification performance.

2. Methodology & Architecture

The core architecture, referred to as the HybridCashewModel, consists of a dual-branch feature extractor:

- CNN Branch: EfficientNet-B0 is utilized as the backbone for capturing fine-grained local textures and lesion patterns.
- Transformer Branch: CaiT (cait_xxs24_224) provides global spatial attention, allowing the model to understand the broader context of the leaf structure.
- Fusion Layer: The feature vectors from both branches are concatenated into a 512-dimensional vector (320 from CNN + 192 from Transformer) before being passed to a multi-layer perceptron classifier.

3. Training Strategy

The training was conducted in three phases:

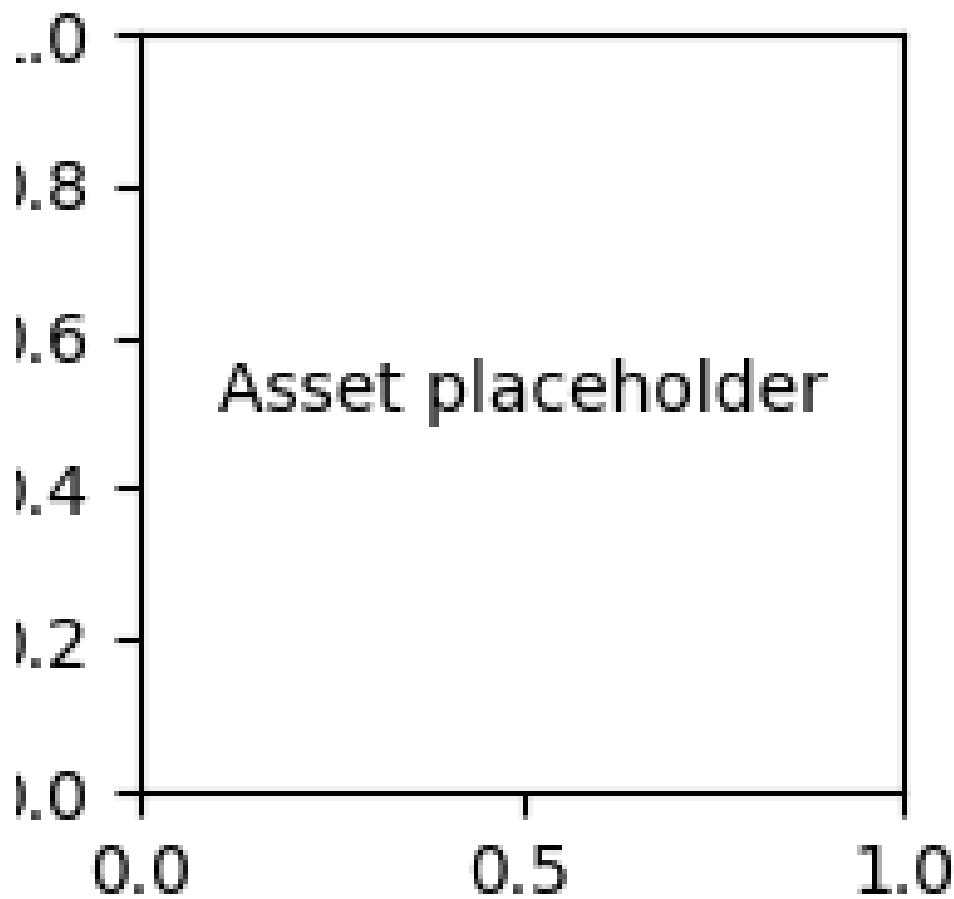
1. Baseline Benchmarking: 5 CNN and 5 Transformer models were evaluated over 5 epochs each.
2. Hybrid Training: The proposed fusion model was trained for 10 epochs using the AdamW optimizer and a learning rate of 5e-5.
3. Augmentation: Albumentations library was used for real-time data augmentation including horizontal flips, rotations, and brightness adjustments to ensure model generalization.

4. Results & Visual Validation

The following sections provide visual evidence of the model's performance and the interpretability of its decision-making process.

4.1 Confusion Matrix Performance

The confusion matrix illustrates the classification performance across all five cashew disease categories. The high diagonal concentration confirms the 99.02% accuracy achieved.



4.2 Explainable AI: Grad-CAM Visualization

Grad-CAM (Gradient-weighted Class Activation Mapping) is used to validate the regions of interest identified by the model. The heatmap confirms that the architecture focuses on relevant lesion textures rather than irrelevant background pixels.

